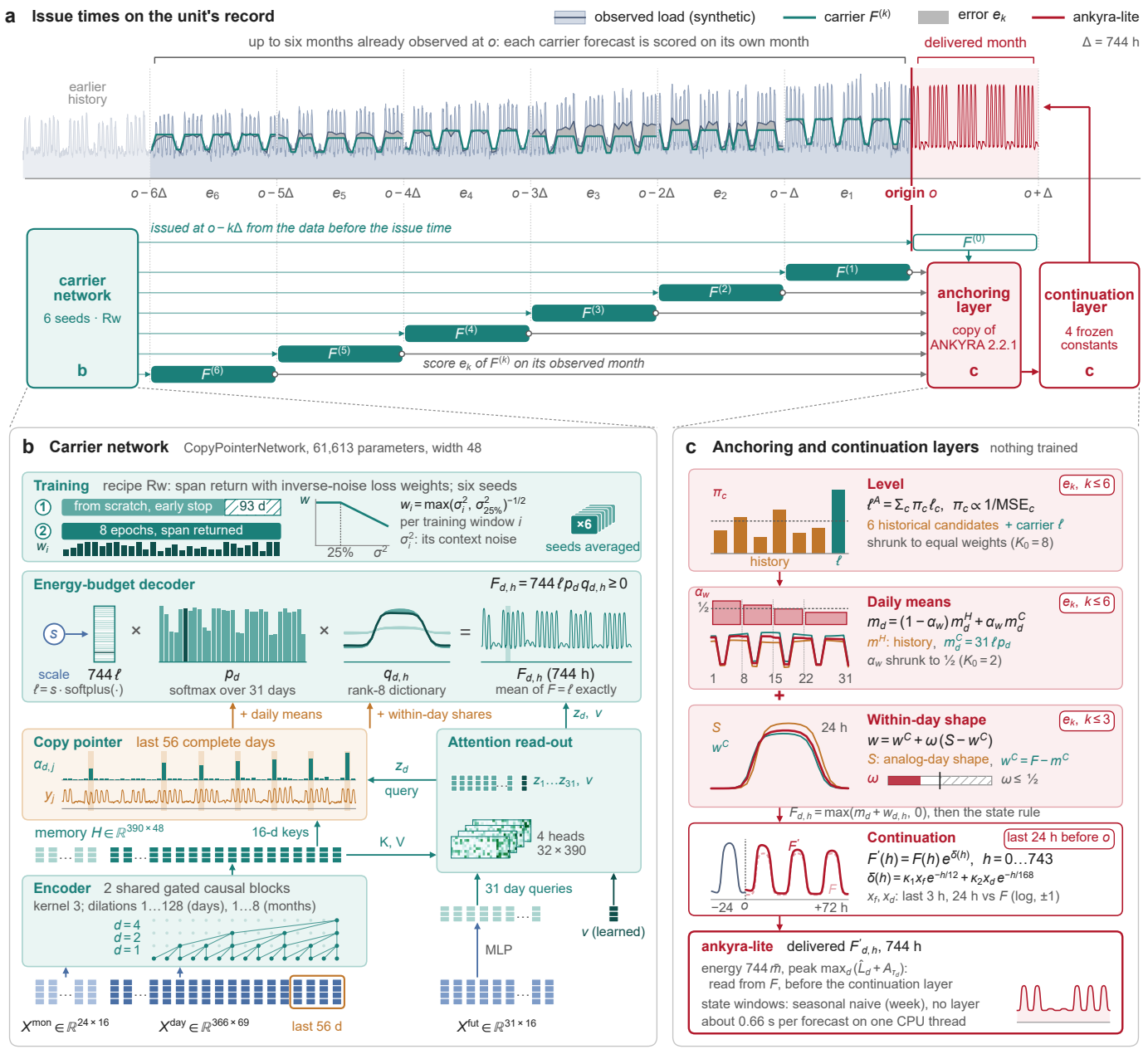

a Issue times on the unit's record
observed load (synthetic)
carrier $F^{(k)}$
error e_k
ankyra-lite
up to six months already observed at o: each carrier forecast is scored on its own month
delivered month
Δ = 744 h
earlier history
o − 6Δ
e6
o − 5Δ
e5
o − 4Δ
e4
o − 3Δ
e3
o − 2Δ
e2
o − Δ
e1
origin o
o + Δ
issued at o − kΔ from the data before the issue time
carrier network
6 seeds · Rw
b
$F^{(0)}$
$F^{(1)}$
$F^{(2)}$
$F^{(3)}$
$F^{(4)}$
$F^{(5)}$
$F^{(6)}$
score e_k of $F^{(k)}$ on its observed month
anchoring layer
copy of ANKYRA 2.2.1
c
continuation layer
4 frozen constants
c
b Carrier network
CopyPointerNetwork, 61,613 parameters, width 48
Training
recipe Rw: span return with inverse-noise loss weights; six seeds
from scratch, early stop
93 d
8 epochs, span returned
w_i
w
25%
σ^2
$w_i = \max(\sigma_i^2, \sigma_{25\%}^2)^{-1/2}$
per training window i
σ_i^2: its context noise
×6
seeds averaged
Energy-budget decoder
$F_{d,h} = 744\,\ell\,p_d\,q_{d,h} \geq 0$
S
scale
744 ℓ
ℓ = s · softplus(·)
p_d
softmax over 31 days
$q_{d,h}$
rank-8 dictionary
$F_{d,h}$ (744 h)
mean of F = ℓ exactly
+ daily means
+ within-day shares
z_d, v
Copy pointer
last 56 complete days
$\alpha_{d,j}$
y_j
z_d
query
Attention read-out
$z_1 \ldots z_{31}$, v
4 heads
32 × 390
memory $H \in \mathbb{R}^{390\times 48}$
16-d keys
K, V
Encoder
2 shared gated causal blocks
kernel 3; dilations 1…128 (days), 1…8 (months)
d = 4
d = 2
d = 1
31 day queries
MLP
v (learned)
$X^{mon} \in \mathbb{R}^{24\times 16}$
$X^{day} \in \mathbb{R}^{366\times 69}$
last 56 d
$X^{fut} \in \mathbb{R}^{31\times 16}$
c Anchoring and continuation layers
nothing trained
π_c
history
ℓ
Level
e_k, k ≤ 6
$\ell^A = \Sigma_c\,\pi_c\,\ell_c$, $\pi_c \propto 1/\mathrm{MSE}_c$
6 historical candidates + carrier ℓ
shrunk to equal weights (K_0 = 8)
α_w
½
1
8
15
22
31
Daily means
e_k, k ≤ 6
$m_d = (1-\alpha_w)\,m_d^H + \alpha_w\,m_d^C$
m^H: history, $m_d^C = 31\,\ell\,p_d$
α_w shrunk to ½ (K_0 = 2)
S
w^C
24 h
Within-day shape
e_k, k ≤ 3
$w = w^C + \omega\,(S - w^C)$
S: analog-day shape, $w^C = F - m^C$
ω
ω ≤ ½
$F_{d,h} = \max(m_d + w_{d,h}, 0)$, then the state rule
F′
F
−24
o
+72 h
Continuation
last 24 h before o
$F'(h) = F(h)\,e^{\delta(h)}$, h = 0…743
$\delta(h) = \kappa_1 x_f e^{-h/12} + \kappa_2 x_d e^{-h/168}$
x_f, x_d: last 3 h, 24 h vs F (log, ±1)
ankyra-lite
delivered $F'_{d,h}$, 744 h
energy 744 m̄, peak $\max_d(\hat{L}_d + A_{\tau_d})$:
read from F, before the continuation layer
state windows: seasonal naive (week), no layer
about 0.66 s per forecast on one CPU thread